

Objectives

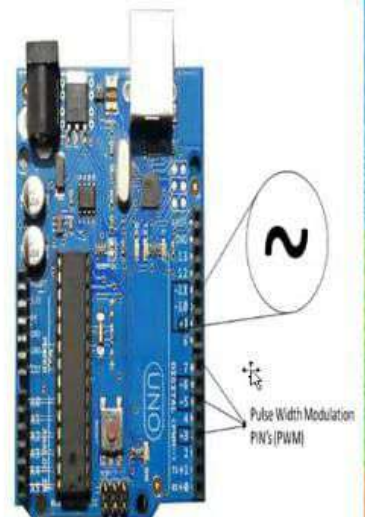
Define the Pulse Width Modulation (PWM)

Control the Pulse Width Modulation

Use the PWM with electric elements

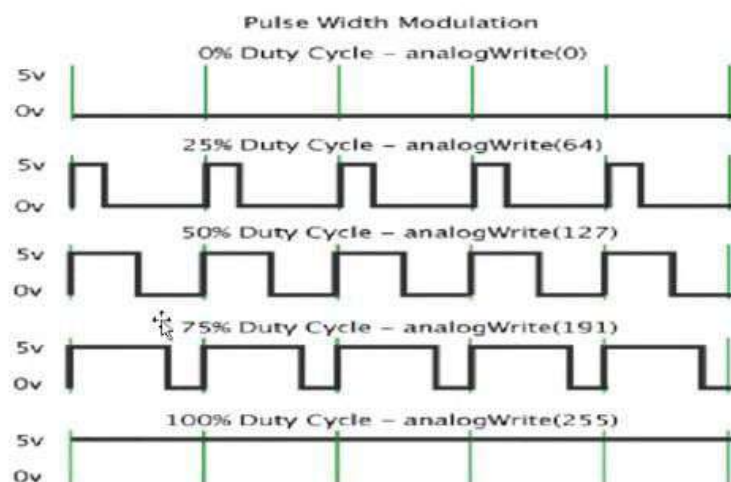
Pulse Width Modulation (PWM)

- Pulse Width Modulation, or PWM, is a technique for getting analog results with digital means.
- Digital control is used to create a square wave, a signal switched between on and off. This on-off pattern can simulate voltages in between full on (5 Volts) and off (0 Volts) by changing the portion of the time the signal spends on versus the time that the signal spends off.
- The duration of "on time" is called the pulse width. To get varying analog values, you change or modulate, that pulse width. If you repeat this on-off pattern fast enough with an LED, for example, the result is as if the signal is a steady voltage between 0 and 5v controlling the brightness of the LED."



Pulse Width Modulation (PWM)

As PWM is how many square waves you are modulating. We can control them by changing the value of square waves. On the figure below you can see the square waves and how to control them with an `analogWrite()` command.



Pulse Width Modulation (PWM)

$$\text{Duty Cycle} = \frac{T_{\text{on}}}{T_{\text{on}} + T_{\text{off}}} \times 100$$

V_{out} (voltage received by the LED) = Duty cycle $\times V_{\text{max}}$ ($V_{\text{max}} = 5\text{V}$).

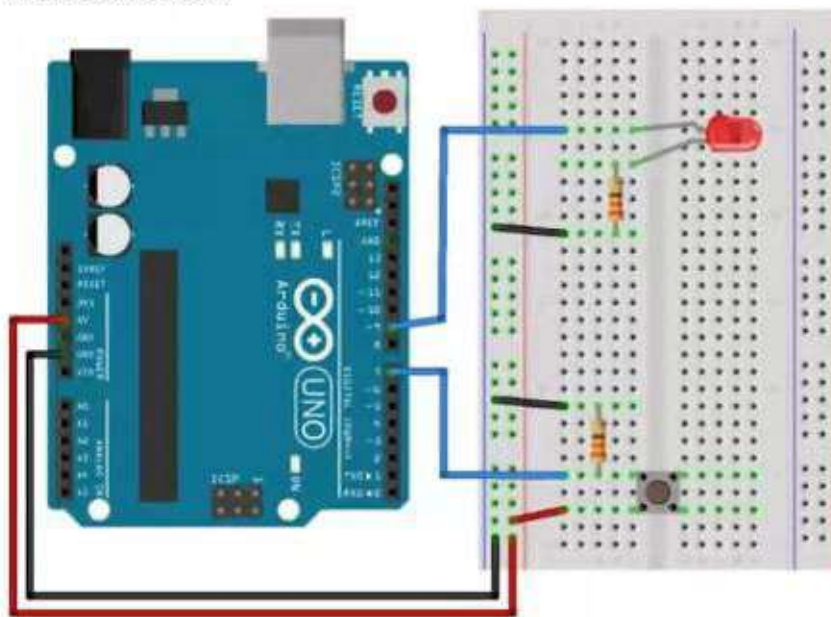
PWM reading value are in the range 0 – 255 ($2^8 = 256$)

Arduino is based on 8 bits microcontrollers, this means they have 8 bits registers) and the first value is for storing the length of the reading values.

- `analogWrite(0)` = 0% duty cycle (0V)
- `analogWrite(63 or 64)` = 25% duty cycle – Low bright (1.25V)
- `analogWrite(126 or 127)` = 50% duty cycle – Medium bright (2.5V)
- `analogWrite(188 or 191)` = 75% duty cycle – High bright (3.75V)
- `analogWrite(255)` = 100% duty cycle – Full bright (5V)

Board connection

Board connection:



sketch_jul23a.ino

```
3 void setup() {
4   // put your setup code here, to run once:
5   pinMode(led,OUTPUT);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  analogWrite(led,63);
11  delay(1000);
12
13  analogWrite(led,126);
14  delay(1000);
15
16  analogWrite(led,191);
17  delay(1000);
18
19  analogWrite(led,53);
20  delay(1000);
21
22  analogWrite(led,63);
23  delay(1000);
24
```

```
3 void setup() {
4   // put your setup code here, to run once:
5   pinMode(led,OUTPUT);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  analogWrite(led,63); //25% duty cycle --> Vled=1.25V (LOW BRIGHTNESS)
11  delay(1000);
12
13  analogWrite(led,126); //50% DUTY CYCLE -->Vled=2.5V(MEDIUM BRIGHTNESS)
14  delay(1000);
15
16  analogWrite(led,191); //75% DUTY CYCLE -->Vled=3.75V(HIGH BRIGHTNESS)
17  delay(1000);
18
19  analogWrite(led,255); //100% DUTY CYCLE -->Vled=5V(FULL BRIGHTNESS)
20  delay(1000);
21
22  analogWrite(led,0);
23  delay(1000);
24
```



```
Sketch_ju23a.ino | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
Sketch_ju23a.ino
1 int led=9; //to simulate a PWM we connect the led
2
3 void setup() {
4   // put your setup code here, to run once:
5   pinMode(led,OUTPUT);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  analogWrite(led,63); //25% duty cycle --> Vled=1.1
11  delay(1000);
12
13  analogWrite(led,126); //50% DUTY CYCLE -->Vled=2.2
14  delay(1000);
15
16  analogWrite(led,191); //75% DUTY CYCLE -->Vled=3.3
17  delay(1000);
18
19  analogWrite(led,255); //100% DUTY CYCLE -->Vled=5V
20  delay(1000);

```

Sketch uses 1166 bytes (3%) of program storage space.

```
Sketch_ju23b.ino | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
Sketch_ju23b.ino
1 int led=9;
2 int power=0;
3
4 void setup() {
5   // put your setup code here, to run once:
6   pinMode(led,OUTPUT);
7 }
8
9 void loop() {
10  // put your main code here, to run repeatedly:
11  for(int i =0;i<4;i++){
12    power=power+63;
13  }
14 }
15 }
16

```

Activate Windows
Go to Settings to activate Windows.


```
sketch_ju73a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_ju73a.ino
1 int led=9; //to simulate a PWM we connect the led
2
3 void setup() {
4   // put your setup code here, to run once:
5   pinMode(led,OUTPUT);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  analogWrite(led,63); //25% duty cycle --> Vled=1.1
11  delay(1000);
12
13  analogWrite(led,126); //50% DUTY CYCLE -->Vled=2.2
14  delay(1000);
15
16  analogWrite(led,191); //75% DUTY CYCLE -->Vled=3.3
17  delay(1000);
18
19  analogWrite(led,255); //100% DUTY CYCLE -->Vled=5V
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